Vesper Project Roadmap

This document summarizes the implementation roadmap for the AI-Powered Alcohol Label Verification App.

Project Goal

Build a decision-support prototype that compares application data against an uploaded alcohol label using OCR. The application assists compliance agents and does not make legal approval decisions.

Core Workflow

1. Enter expected label data.
2. Upload label image.
3. OCR extracts text.
4. Normalize extracted values.
5. Compare against expected values.
6. Display Match, Possible Match, Mismatch, or Unable to Determine.
7. Human makes the final decision.

Prototype Scope

Support distilled spirits only. Validate Brand Name, Class/Type, ABV, Net Contents, and Government Warning.

Recommended Technology

Frontend: Streamlit
Backend: Python
OCR: Local OCR (EasyOCR/PaddleOCR preferred) with optional Azure provider.
Comparison: Deterministic rules plus fuzzy matching where appropriate.

Architecture

Upload → Image preprocessing → OCR → Field extraction → Normalization → Comparison → Results.

Validation

Use exact matching for government warning wording and normalized matching for brand/class names. Convert proof and ABV, normalize units, and flag uncertain matches for manual review.

User Experience

Single-page interface, clear upload controls, processing time display, confidence scores when available, and helpful error messages.

Testing

Create sample labels covering perfect matches, mismatches, capitalization differences, missing warnings, low quality images, rotated labels, and unreadable labels.

Repository

Include source code, README, requirements.txt, tests, sample labels, assumptions, limitations, and trade-offs.

Design Principles

Fast local processing, human-in-the-loop, modular architecture, deterministic validation where possible, AI only where it adds value.